



Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories — exploring spending patterns, customer segments, product preferences, and subscription behavior to guide strategic decisions.

Dataset at a Glance

3,900

Transactions

18

Columns

\$59.76

Avg. Purchase

3.75

Avg. Rating

Key Feature Groups

- **Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item, Category, Amount, Season, Size, Color
- **Behavior:** Discounts, Promo Codes, Frequency, Ratings, Shipping

❏ 37 missing values in Review Rating — imputed using median per product category.

Data Preparation in Python

01

Load & Explore

Imported dataset with pandas; used `df.info()` and `.describe()` for structure and summary statistics.

02

Clean & Standardize

Imputed missing Review Ratings; renamed columns to **snake_case**; dropped redundant `promo_code_used` column.

03

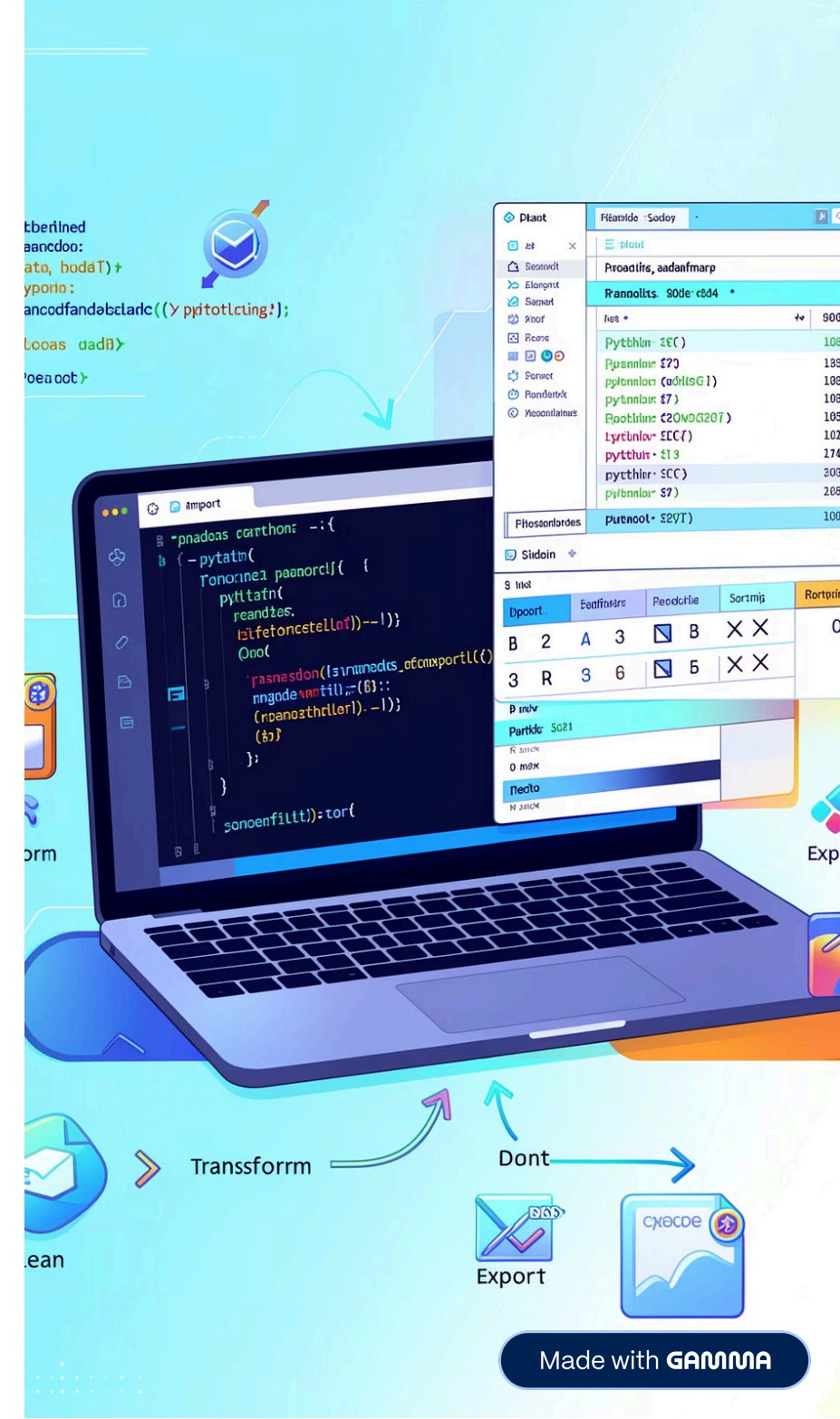
Feature Engineering

Created `age_group` bins and `purchase_frequency_days` from purchase data.

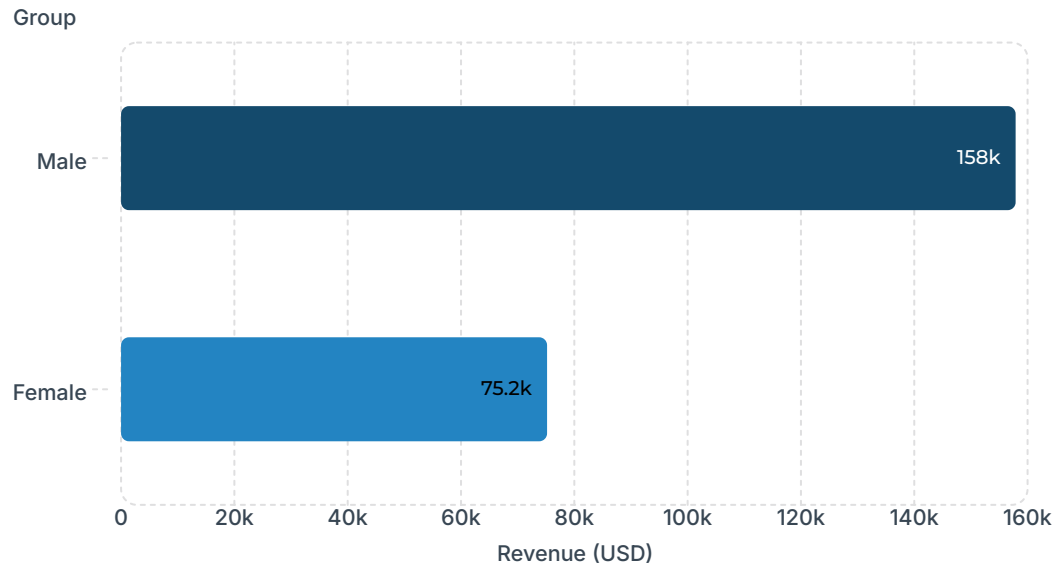
04

Database Integration

Connected Python to **PostgreSQL** and loaded the cleaned DataFrame for SQL analysis.



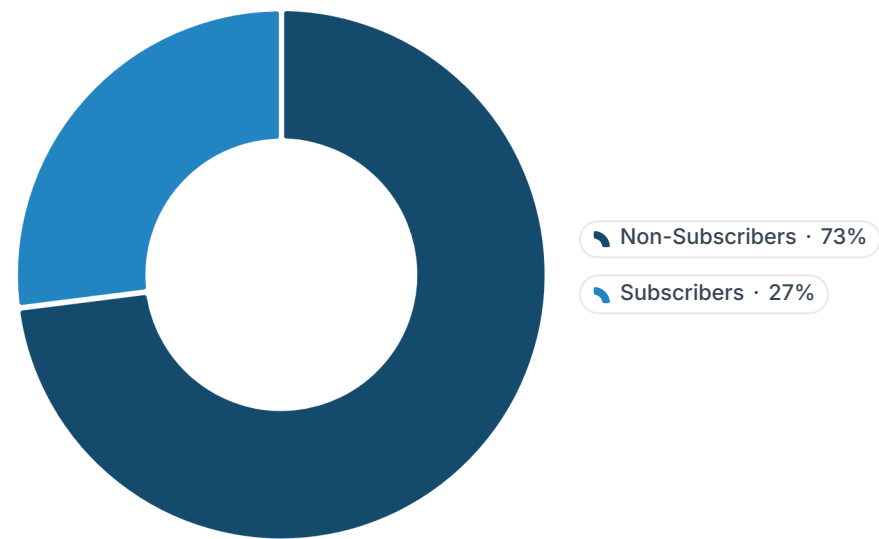
Revenue & Spending Insights



Key Findings

- **Male customers** generated \$157,890 vs. **Female** at \$75,191 in total revenue.
- **Express shipping** users averaged \$60.48 vs. \$58.46 for Standard — a small but notable premium.
- **839 customers** used discounts yet still spent above the average purchase amount.

Subscription & Loyalty Behavior



Subscribers vs. Non-Subscribers

Status	Customer s	Avg. Spend	Total Revenue
Subscriber s	1,053	\$59.49	\$62,645
Non- Subscriber s	2,847	\$59.87	\$170,436

Only **27%** are subscribers. Repeat buyers (>5 purchases): **958 subscribed** vs. 2,518 non-subscribed — a clear loyalty opportunity.

Customer Segmentation



Segment Breakdown

Customers classified by previous purchase count. The vast majority are already loyal — focus shifts to retention and upsell.

Loyal

3,116 customers — the dominant segment

Returning

701 customers — re-engaged buyers

New

83 customers — acquisition pipeline

Product Performance

Top 5 Products by Rating

Product	Avg. Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

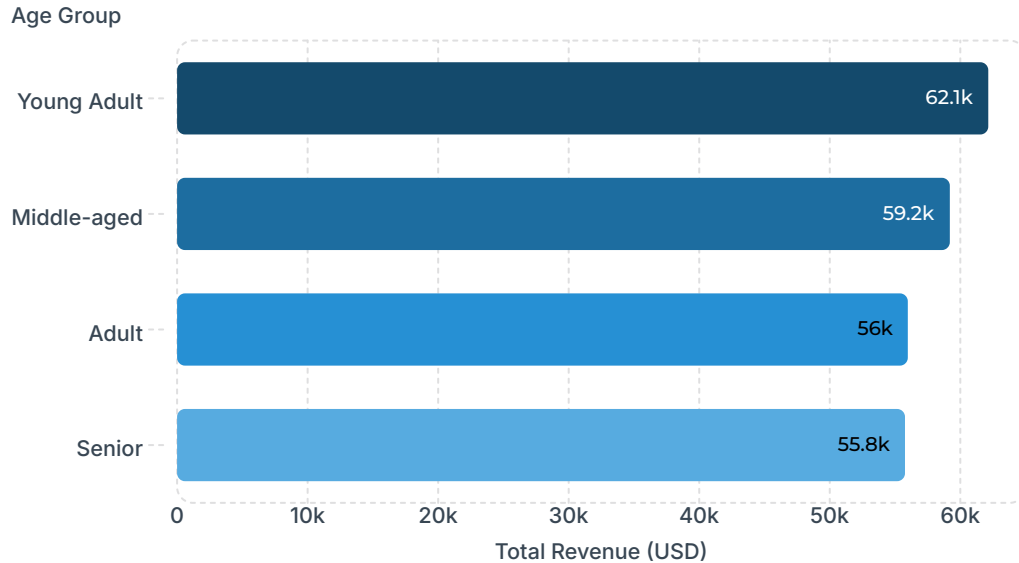
Top Products by Category

Category	#1 Product	Orders
Accessories	Jewelry	171
Clothing	Blouse	171
Footwear	Sandals	160
Outerwear	Jacket	163

Most Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%

Revenue by Age Group



Age-Driven Insights

Young Adults lead revenue at \$62,143, followed closely by Middle-aged at \$59,197. Revenue is relatively balanced across all age groups, suggesting broad market appeal.

i Targeted campaigns toward Young Adults and Middle-aged segments offer the highest revenue upside.

Power BI Dashboard



An interactive dashboard with filters for Subscription Status, Gender, Category, and Shipping Type — featuring KPI cards, a subscription donut chart, and revenue/sales breakdowns by category and age group.



Business Recommendations



Boost Subscriptions

Promote exclusive benefits to convert the **73%** non-subscriber base.



Loyalty Programs

Reward repeat buyers to accelerate movement into the **Loyal** segment.



Review Discount Policy

Balance sales boosts with margin control — especially for Hat, Sneakers, and Coat.



Targeted Marketing

Focus on **Young Adults** and **express-shipping** users for highest revenue impact.



Product Positioning

Highlight top-rated (Gloves, Sandals) and best-selling products in campaigns.